

User Manual

Hospital Real Seismic Monitoring System

Version 1.0 — production branch

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1. Introduction

This document describes the complete operation of the accelerometer data acquisition system installed at the Hospital Real (University of Granada). The system continuously records the acceleration measured by several sensors connected to a National Instruments device, stores the data, and lets you analyse it in the frequency domain.

The manual is written for the people who operate the equipment day to day. No programming knowledge is required: every routine task is carried out from the graphical interface or from the .bat files included in the project.

1.1. What the system does

- Captures the signal from the accelerometers through the National Instruments DAQ, at the configured sampling rate.
- Reduces the sampling rate (decimation) so that the stored volume of data stays manageable without losing the content of interest.
- Stores the data, with its UNIX timestamp, in a local SQLite database that acts as a fast intermediate buffer.
- Periodically copies that data to a MongoDB database for historical archiving, and frees the local database.
- Lets you plot the recorded signal and compute the power spectral density (PSD) and the singular value decomposition (SVD).

1.2. Terms used in this manual

Term	Meaning
DAQ	National Instruments data acquisition equipment (cDAQ chassis with NI 9230 modules).
Channel / sensor	Each accelerometer input. It is identified by two digits: the first is the module number and the second is the channel within that module. For example, 21 is module 2, channel 1.
Decimation	Reduction of the sampling rate by an integer factor, with an antialiasing filter applied beforehand.
Buffer	Intermediate memory. Here it refers both to the DAQ buffer, measured in samples, and to the local SQLite database.
Backup	Transfer of the data from SQLite to MongoDB.
PSD	Power spectral density. Shows how the energy of the signal is distributed across frequencies.
SVD	Singular value decomposition of the PSD matrix. Used to identify the vibration modes of the structure.

2. Prerequisites

2.1. Hardware

- National Instruments cDAQ9185 or cDAQ9189 chassis, connected to the PC over the network or by USB.

- One or more NI 9230 modules, with 3 channels each.
- Accelerometers connected to the channels. The modules must occupy the chassis slots in ascending order.
- A PC running Windows 10 or 11, with enough free disk space for the local database.

2.2. Software

- The NI-DAQmx driver, tested with version 24.3, together with the NI MAX tool.
- Python 3.10.
- MongoDB (optional). If it is not available the system still works and keeps everything in SQLite.

Note: Before installing anything, check in NI MAX that the chassis is listed and that you can read the channels. If NI MAX cannot see the equipment, the acquisition software will not see it either.

3. Installation

The installation is done once per computer. Every command is run from a command prompt located in the root folder of the project.

3.1. Getting the code

The repository has two reference branches: production holds the deployed and verified code, and main holds work in progress. On a production machine always use production.

```
git clone
cd Codigos_NI
git checkout production
```

3.2. Creating the virtual environment and installing the dependencies

```
python -m venv .venv
.venv\Scripts\activate
pip install --upgrade pip
pip install -r requirements.txt
deactivate
```

The .venv folder is created inside the project and holds every required library. It must not be copied between machines: if the project is moved to another PC, repeat this step there.

3.3. Checking the installation

Run the automatic check. It verifies that every module imports, that SETUP.txt is parsed correctly, and that the database tables can be created. It does not use the DAQ or MongoDB, so it can be run with the equipment disconnected.

```
.venv\Scripts\activate
python test_smoke.py
```

The expected result is the message “OK: smoke test passed”. If an error appears, check that the active branch is production and that the dependency installation finished without failures.

4. File structure

Every program must be run from the root folder of the project, because the paths to SETUP.txt and to the result files are relative to it.

Item	Description
SETUP.txt	Configuration file. It is the only file the user needs to modify.
RUNGUI.bat	Opens the graphical control panel.
RUNMAIN.bat	Starts the acquisition without a graphical interface.
test_smoke.py	Installation check.
requirements.txt	List of the required Python libraries.
src/	Application code: acquisition, databases and interfaces.
control/	Utility that inspects the MongoDB database.
doc/	This manual, in DOCX and PDF format.
last_seconds.json	Working file holding the last seconds read from the database.
psd_results.json	Working file holding the PSD and SVD results drawn by the checking interface.

4.1. Modules in the src folder

File	Purpose
main.py	Entry point. Loads the configuration, prepares the databases, launches the acquisition, and handles stopping and periodic restarting.
data_acquisition.py	Starts the three parallel processes: capture, processing and backup.
capture_data.py	Reads the samples from the DAQ and sends them to the processing queue.
process_data.py	Decimates the signals, assigns the timestamps, and groups the data before writing it.
data_handling.py	Writes to SQLite and performs the SQLite to MongoDB backup.
setup_database.py	Creates the tables and indexes, and manages the MongoDB connection.
load_setup_data.py	Reads and parses SETUP.txt.
utils.py	Exports recent data and computes the PSD and the SVD.
GUI.py	Acquisition control panel.
GUI_check.py	PSD and SVD viewer.

5. Configuration: the SETUP.txt file

SETUP.txt defines how the system behaves. It can be edited with any text editor, but it is best edited from the control panel, which converts the data types and preserves the comments in the file. Each line has the form “key: value # comment”. Lines starting with # are comments and are ignored.

5.1. Capture parameters

Parameter	Values	Description
total_capture_time	0 or seconds	Total duration of the capture. With 0 the capture runs until it is stopped manually. If a value greater than 0 is given, the periodic restart is disabled automatically.
backup_time	0 or seconds	Interval between backups from SQLite to MongoDB. With 0 no backups are made and all the data stays in SQLite.
restart_time	0, 10s, 1h, 1d, 1w	Interval after which the acquisition stops and starts again. Restarting periodically prevents timing drift from building up in very long captures. With 0 it never restarts.
debug	true / false	Prints detailed information about every packet to the console. Useful for diagnosis, but it produces a lot of output and slows execution down.

5.2. DAQ parameters

Parameter	Values	Description
deviceName	Text	Name of the device exactly as it appears in NI MAX, for example cDAQ1 or cDAQ9185-21DA9E1. It must match exactly.
original_rate	2 to 24000 Hz	Original sampling rate of the DAQ. High values combined with decimation are recommended.
buffer_size	Samples	Size of the DAQ buffer. The equipment delivers data when the buffer fills: at 1600 Hz with a 1600-sample buffer, it delivers one block per second.
decimation_factor	Integer	Divisor applied to the sampling rate. The stored rate is original_rate divided by this factor.
daq_num_modules	Integer	Number of modules connected to the chassis. They must occupy the slots in ascending order.
daq_ch_per_module	Integer	Channels per module, normally 3 on the NI 9230.
unused_ch	nan or list	Channels that are not used. The value nan means every channel is used. To exclude channels, list their two-digit identifiers separated by commas, for example: 21, 22.

5.3. Sensor parameters

Parameter	Unit	Description
min_val	m/s ²	Minimum value expected in the measurement. Take it from the datasheet of the accelerometer.
max_val	m/s ²	Maximum value expected in the measurement.
sensitivity	V/g	Sensitivity of the accelerometer. A wrong value mis-scales the whole recorded signal.

Note: *min_val*, *max_val* and *sensitivity* depend on the sensor model installed. If an accelerometer is replaced with a different model, these three values must be updated before the next capture.

5.4. Database parameters

Parameter	Description
db_path	Folder where the SQLite file is created. If it does not exist, the program creates it.
sqlite_name	Name of the SQLite file, without extension. Changing it is equivalent to starting a new database.
db_host	Address of the MongoDB server, normally localhost.
db_port	MongoDB port, 27017 by default.
db_backup_name	Name of the MongoDB database used for historical storage.

6. Control panel

This is the usual way to operate the system. Open it with RUNGUI.bat, or by running “python src\GUI.py” from the root folder. A loading screen appears briefly and then the maximised window opens.

6.1. Configuration area

The upper part shows the SETUP.txt parameters grouped into sections: capture, DAQ, sensors, buffer database and backup database. If the list does not fit on screen, scroll it with the side bar or the mouse wheel.

- Each parameter shows its unit on the right and a pop-up description when the cursor is placed over the field.
- Critical fields are locked, marked with a padlock icon. To change one, click the padlock; the icon switches to open and the field becomes editable. This lock prevents accidental changes to parameters that affect the consistency of data already recorded.
- The frequency_rate field is informative: it shows the resulting rate after decimation and updates when the factor changes.
- The “Save Changes” button writes the values into SETUP.txt, keeping the original comments of the file.

Note: Saving the changes does not affect an acquisition that is already running. To apply new parameters, stop the acquisition and start it again.

6.2. Plot area

Control	Purpose
Plot Once	Draws the last few seconds recorded in the local database, once.
Auto-update	Redraws the plot automatically every few seconds while the box is ticked.
Find Sensors	Queries the database and refreshes the list of available sensors, with one checkbox per sensor to show or hide it.
Time Lag	Shows the delay between the last stored sample and the current time. A value that keeps growing means the computer is not keeping up with the acquisition.

6.3. Control buttons

Button	Purpose
Run Acquisition	Saves the configuration and opens a new console where the acquisition runs. The button stays disabled while the process is alive, to prevent two simultaneous acquisitions.
Stop Acquisition	Sends the stop signal. The acquisition detects it within one second and shuts the processes down cleanly.
Check Data	Opens the PSD and SVD checking interface in a separate window.

6.4. Interface console

The text box below the configuration shows the initial instructions and the result of every action: configuration saved, acquisition started, connection errors. It is worth reading after each operation.

6.5. Closing the window

When the control panel is closed, if there are unsaved changes the program asks whether to keep them. If you answer no, SETUP.txt is restored to the state it had when the application was opened. Closing the panel does not stop an acquisition that is running.

7. Checking interface (PSD and SVD)

Open it with the “Check Data” button of the control panel, or by running “python src\GUI_check.py” from the root folder. It reads the last 30 seconds from the local database, computes the power spectral density matrix and its singular value decomposition, and plots the results.

7.1. Controls

Control	Purpose
Plot Once	Reads the data, recomputes the PSD and the SVD, and refreshes the plot once.
Auto-update	Repeats the computation continuously. While it is active, “Plot Once” is disabled.
Min / Max Frequency	Limit the frequency range shown, in hertz. Press “Refresh” to apply the new limits.
Log Scale	Switches the vertical axis between linear and logarithmic. The logarithmic scale makes low-amplitude peaks easier to see.
Show PSDs Overlay	Overlays the PSDs of the selected sensors in a single plot.
Show PSDs Mosaic	Shows one PSD per sensor in stacked plots. Hovering over a curve displays the frequency and the value at that point.
Show SVD	Plots the singular values against frequency. The peaks of the first singular value correspond to the natural frequencies of the structure.

7.2. Sensor selection

In the overlay view a row of checkboxes appears, one per sensor. Unticking a sensor removes it from the plot without affecting the stored data. The selection is kept between refreshes.

Note: If a warning says *psd_results.json* was not found, press “Plot Once” once: that file is created by the first computation.

8. Acquisition without the graphical interface

For unattended captures, or to start the system after rebooting the computer, run RUNMAIN.bat, or the following commands:

```
.venv\Scripts\activate  
python src\main.py
```

The console shows the path of the database, the result of the SQLite and MongoDB setup and, if debug is enabled, information about every captured packet.

8.1. Stopping the acquisition

- Press ENTER in the acquisition console.
- Or press “Stop Acquisition” in the control panel. This option creates a STOP.txt file in the root folder, which the acquisition checks once per second; when it finds the file it stops and deletes it.

Both methods shut the capture, processing and backup processes down cleanly, and close the MongoDB connection. If the console does not close by itself after a few seconds, you can close it manually.

Note: Do not close the acquisition console with the window cross while a backup is running: the processes would be interrupted without finishing the write. Always use ENTER or the stop button.

9. Where the data is stored

9.1. Local database (SQLite)

It is created at the path given by db_path and sqlite_name. It runs in WAL mode, which allows reading and writing at the same time. It holds three tables:

Table	Content
timestamps	UNIX timestamp of each sample.
sensors	Identifier number of each sensor.
accelerations	Acceleration value, linked to one row of timestamps and one of sensors.

9.2. Historical database (MongoDB)

Every backup_time seconds the data in SQLite is moved to the database named in db_backup_name and deleted from the local database. In MongoDB each document groups all the readings of the same instant in a sensor_data field. If MongoDB is unreachable at start-up, the system disables the backups, reports it on the console, and carries on recording in SQLite.

Note: With backups disabled (backup_time equal to 0) the local database grows without bound. On long captures, keep an eye on the free disk space.

10. Maintenance and verification

Task	Command	When
Check the installation	<code>python test_smoke.py</code>	After installing or updating the software, or if something stops working.
Inspect MongoDB	<code>python control/check_mongo_db.py</code>	Periodically. It detects duplicated timestamps and records with no sensor data.
Update the software	<code>git pull</code>	From the production branch, with the acquisition stopped.

It is also worth checking the free disk space before starting long campaigns, recording any change to the sensor configuration in the installation log, and changing `sqlite_name` whenever you want to start a new record without mixing it with the previous one.

11. Troubleshooting

Symptom	Likely cause	What to do
The acquisition does not start and the message mentions the device	The <code>deviceName</code> value does not match the real name of the equipment.	Open NI MAX, copy the exact name of the chassis, and correct it in the control panel.
Messages “[ERROR] Error capturing data” appear	The DAQ is not responding, is disconnected, or another application is using it.	Close any other NI application, check the network or USB connection, and verify the reading in NI MAX.
A warning says MongoDB is not available	The MongoDB service is not running, or the host and port are wrong.	Start the MongoDB service and check <code>db_host</code> and <code>db_port</code> . The acquisition carries on in SQLite meanwhile.
The Time Lag indicator keeps growing	The sampling rate is too high for the computer, or the decimation is not enough.	Increase <code>decimation_factor</code> or reduce <code>original_rate</code> . Turn debug off, since it slows execution down.
The plot is empty	There is no data in the local database yet, or the sensors are unticked.	Press “Find Sensors” and check that an acquisition is running.
A warning says <code>psd_results.json</code> was not found	No computation has been run yet.	Press “Plot Once” in the checking interface.
The acquisition window does not close when the stop button is pressed	The process is finishing a backup that is under way.	Wait a few seconds; if it persists, press ENTER in that console.
Fewer sensors are recorded than are installed	Some channels are excluded in <code>unused_ch</code> , or <code>daq_num_modules</code> does not match the installed hardware.	Check both parameters in the DAQ section of the control panel.

Symptom	Likely cause	What to do
The dependencies fail to install	The Python version is not 3.10, or the virtual environment is not active.	Check “python --version” and that the command prompt shows (.venv) at the start of the line.

12. Good practice

- Always save the configuration before closing the control panel if you want to keep it.
- Stop the acquisition before changing any parameter: changes are not applied while it runs.
- Keep debug off during normal operation and turn it on only for diagnosis.
- Use a restart_time other than zero for campaigns lasting several days: the periodic restart prevents timing drift from building up.
- Change sqlite_name when starting a new campaign, so records from different periods are not mixed.
- On a production machine always work on the production branch. The main branch may contain unverified changes.

Appendix A. Quick reference

Action	Command
Open the control panel	RUNGUI.bat
Start the acquisition without the interface	RUNMAIN.bat
Open the checking interface	python src\GUI_check.py
Check the installation	python test_smoke.py
Inspect MongoDB	python control\check_mongo_db.py
Stop the acquisition	ENTER in the console, or “Stop Acquisition”

Appendix B. Credits

Developed by Emilio Daroch (University of Granada, 2024-2025) for the Hospital Real Seismic Monitoring System, part of the cognitive structural health monitoring platform of the BUILDCHAIN project.